

Multimedia Databases

The Image Medium

Prof. Dr. Michael Granitzer
Prof. Dr. Harald Kosch

Table of Contents

Image

- 1 Basics
- 2 Image Resolution
- 3 Image Organization
- 4 Color reduction / quantization
- 5 Image Formats

Client

Multimedia-Query

Multimedia-Data (no streaming)

Descriptive information

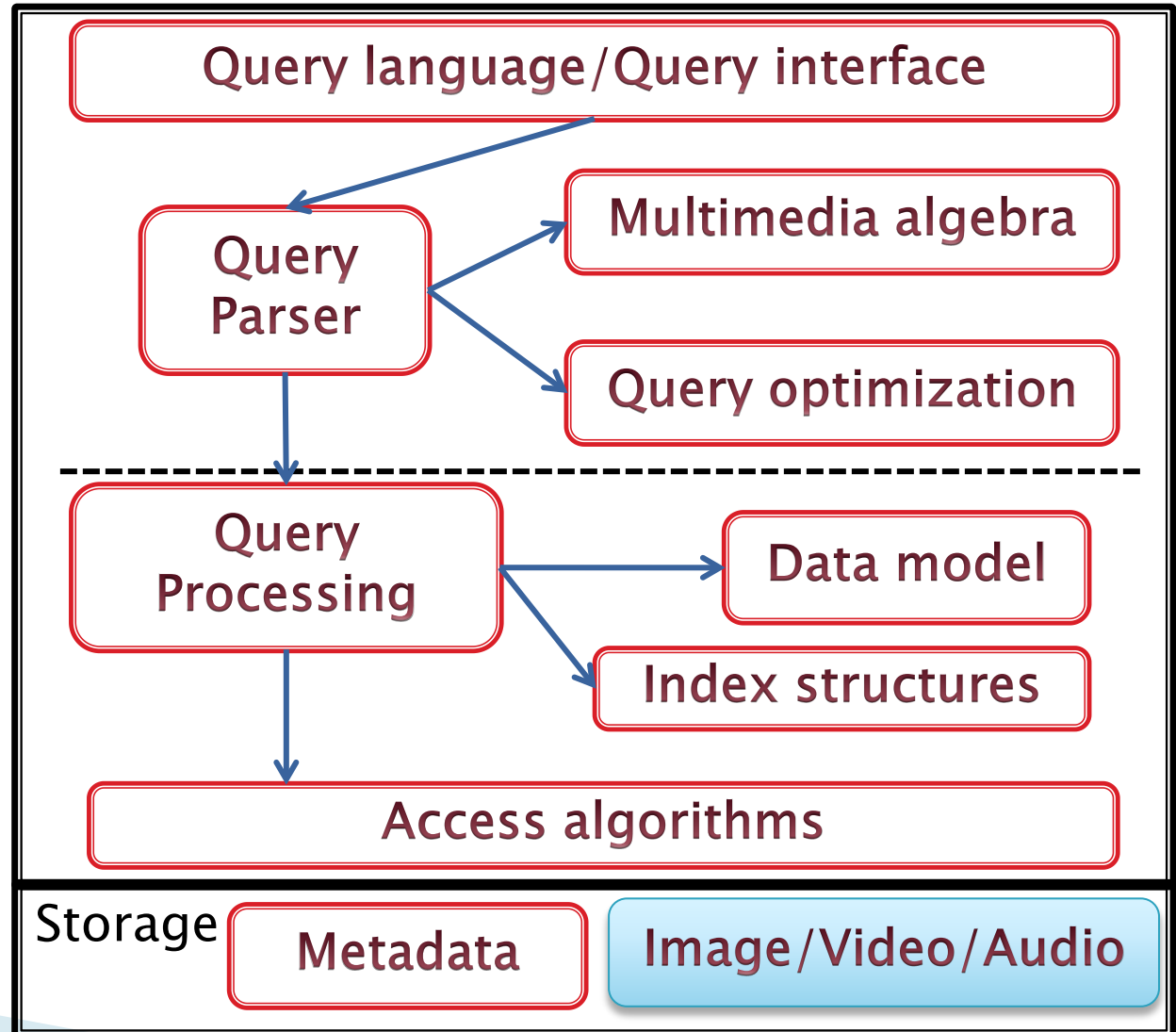


Table of Contents

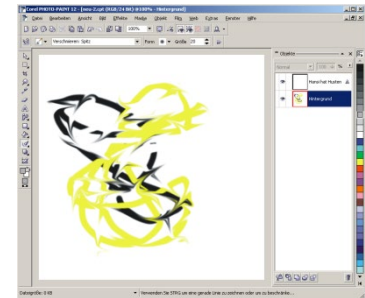
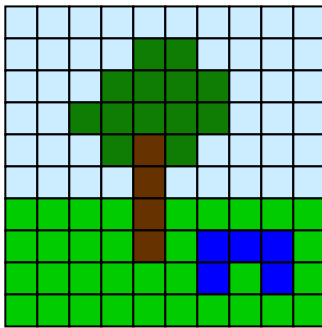
Image

- 1 **Basics**
- 2 Image Resolution
- 3 Image Organization
- 4 Color reduction / quantization
- 5 Image Formats

Basics

Digital Images

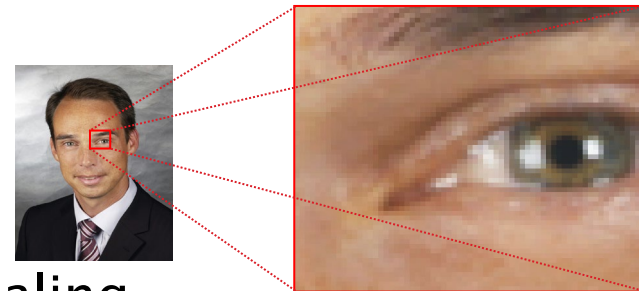
- „Bitmapped Images“: Assignment of color values to single **Image Elements**



- Sources:** scanner, digital camera, drawing programs, ...

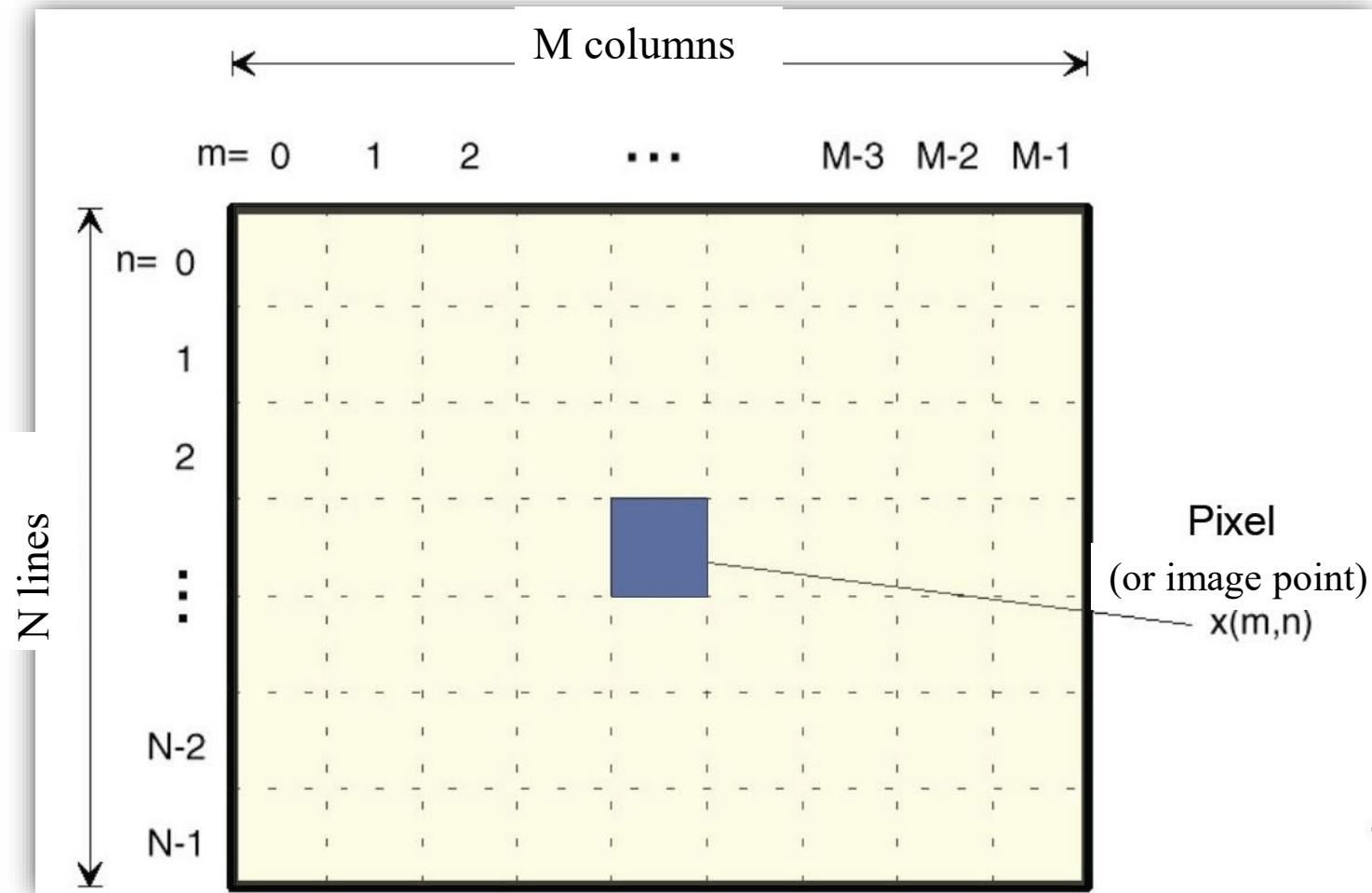
Drawing programs use analogies: brush, spray can, ...

- Disadvantages:** need of memory space, scaling



Basics

Digitale Images II



Basics

Digital Image **Editing** and Image **Processing**

▶ **Editing**

- Means the modification (Improvement, alteration or manipulation) of digital images with specialized software (for example Adobe Photoshop).

▶ **Processing**

- Provides the mathematical algorithms used for editing and professionally modifying/analyzing digital images

Basics

Uses of Digital Image Processing

- ▶ „Image improvement“
 - Image restoration, Retouche
- ▶ Image analysis, Image evaluation
 - Segmentation, analysis of texture
 - Extraction of object contours
- ▶ Image classification, Image recognition, Image sorting

Table of Contents

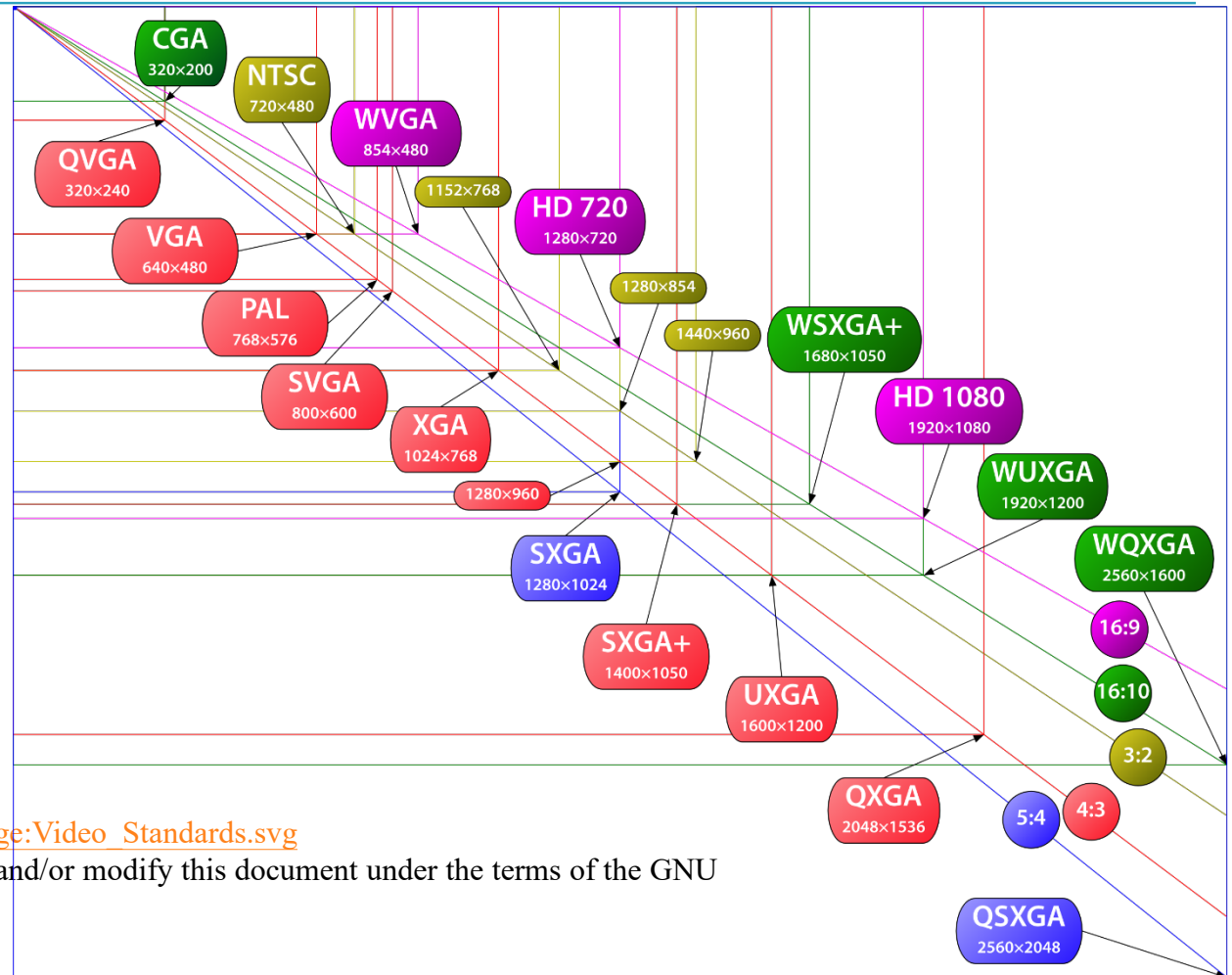
Image

- 1 Basics
- 2 **Image Resolution**
- 3 Image Organization
- 4 Color reduction / quantization
- 5 Image Formats

Device Resolution

- ▶ How accurately is the device/system approximating the image?
- ▶ Printer, Scanner:
 - Relative value in „dots per unit length“, often „dots per inch“ (**dpi**)
 - Desktop Printer 600dpi, Typesetter 1270dpi, Scanner 300–3600dpi, ...
- ▶ Television, Monitors, Digital cameras:
 - Absolute value in Pixel = Pixel Dimension (=Monitor Resolution)
 - PAL TV 768 x 576 px, 17" LCD Monitor 1024 x 768 px, ...
 - The resulting dpi depend on resolution and size of the device

Overview: Pixel Dimension



Source: http://de.wikipedia.org/wiki/Image:Video_Standards.svg

Permission is granted to copy, distribute and/or modify this document under the terms of the GNU Free Documentation License

Image Resolution

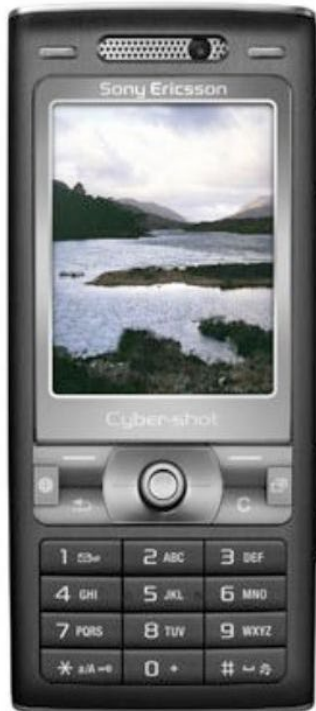
- ▶ The image is basically a **2D Array** of pixels
- ▶ The array has a size in pixels but no physical dimension
- ▶ It follows:
 - The pixel dimension (e.g., 1280x1024) and a physical dimension (19") let one calculate the Image Resolution in **pixels per inch (ppi)**

$$\bullet \text{ *Image Resolution (ppi) = } \frac{\text{Pixel dimension}}{\text{Physical Dimension}}*$$

- ▶ Ex: 19" monitor with a pixel dimension 1280x1024 :

	Pixel	mm	Pixel pro mm	Inch	„ppi“
Width	1280	376	3,4	14,8	86,47
Height	1024	301	3,4	11,85	86,41

Example: Resolution I



K800i

240x320 Pixel (200 ppi)

2" Diagonale

262144 Farben

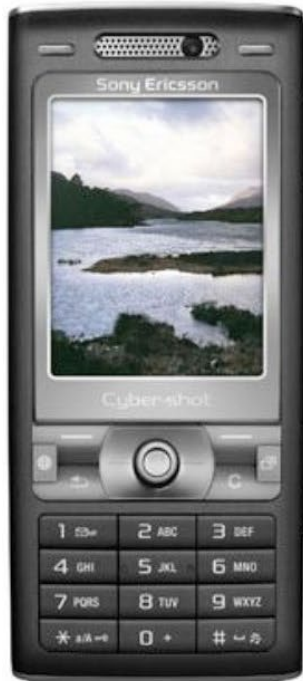
iPhone

320 x 480 Pixel 3.5"
(ca. 160 ppi)

16 Millionen Farben



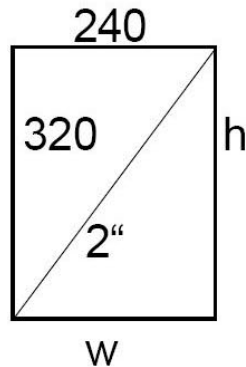
Example: Resolution II



W800i

240x320 Pixel 2" Diagonale

262144 Farben



Width w :

$$w^2 + h^2 = 2^2$$

$$w^2 + \left(\frac{4}{3}w\right)^2 = 4$$

$$w^2 \cdot \left(1 + \frac{16}{9}\right) = 4$$

$$w^2 \cdot \left(\frac{25}{9}\right) = 4$$

$$w^2 = \frac{9 \cdot 4}{25}$$

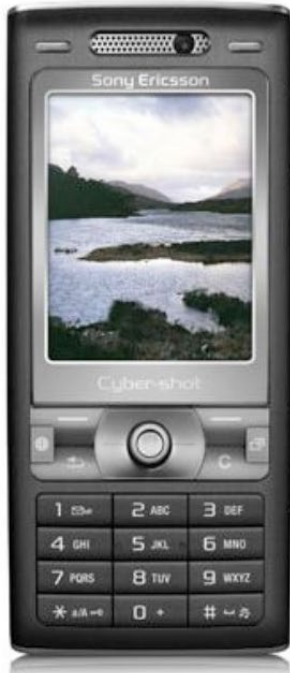
$$w = \frac{6}{5} \text{ inch}$$

Resolution:

$$\frac{6}{5} \text{ inch} \overset{\wedge}{=} 240 \text{ Pixel}$$

$$1 \text{ inch} \overset{\wedge}{=} 200 \text{ Pixel} \\ \Rightarrow 200 \text{ ppi}$$

Example: Resolution III



W800i
240x320 Pixel
262144 Farben

Bit depth:

$$2^b = 262144$$

$$\log_2(2^b) = \log_2(262144)$$

$$b = 18$$

$$\Rightarrow 18/3 = 6 \text{ Bits per channel}$$

Memory requirements:

$$240 \cdot 320 \cdot 18 \text{ bit}$$

$$= 1382400 \text{ Bits} \rightarrow 172800 \text{ Bytes}$$

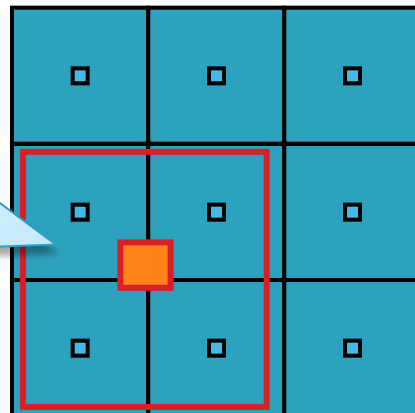
Resolution change

- ▶ If **Image Resolution < Device Resolution: Interpolation** required
 - Leads to **loss of quality**
 - Example → next slides
- ▶ If **Image Resolution > Device Resolution: „Downsampling“** required
 - **Subjective quality** partially better than Device Resolution, because more information is available
 - The technique through which the resolution (finer sampling) of an image is set higher than required by the display device is called **„Oversampling“**
 - Oversampling is useful if good algorithms are used for Downsampling

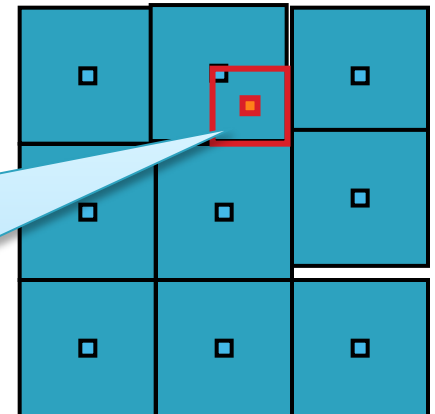
Example: Scaling

- ▶ We want to scale up an image to a factor s
- ▶ This amounts to compute $P'(x, y)$ from $P(x/s, y/s)$
- ▶ Problem: in general x/s and y/s are not integers
- ▶ Therefore: Interpolation required

Pixels become „bigger“ (downscaling) 4->1



Pixels become „smaller“ 1->4



Different Types of Interpolation (downsampling)

▶ Nearest neighbour

- The new Pixel $P'(x,y)$ gets the color of the original Pixel which center is the closest to x/s , y/s

▶ Bilinear Interpolation

- Use the color values of the „covered“ pixels weighted by the size of the respective intersection

▶ Bicubic Interpolation

- Perform no linear Interpolation, use cubic Splines (similar to Bezier Curves) instead for interpolating

Height = 50 Pixel each (enlarged here)

Original



nearest neigh.



bilinear



bikubisch



Table of Contents

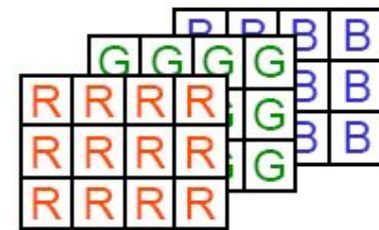
Image

- 1 Basics
- 2 Image Resolution
- 3 **Image Organization**
- 4 Color reduction / quantization
- 5 Image Formats

Pixel arrangement

- Organization of the data in the file or storage system (after transmission)

Example: image of 3 lines of 4 pixels



Pixel Interleaved:

RGBRGBRGBRGBRGBRGBRGBRGBRGBRGBRGBRGB

Line Interleaved:

RRRRGGGGBBBBRRRRGGGGBBBBRRRRGGGGBBBB

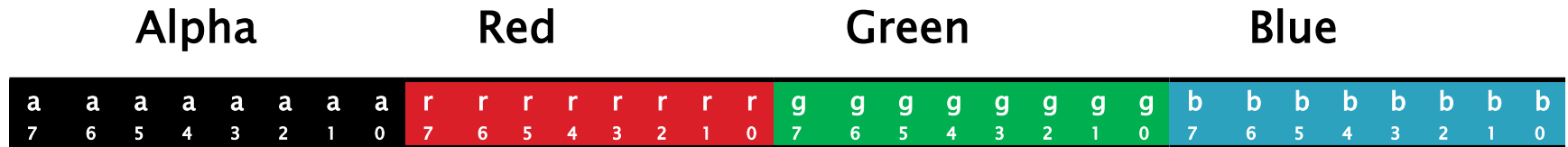
Plane Interleaved:

RRRRRRRRRRRRGGGGGGGGGGGGBBBBBBBBBBBBBB

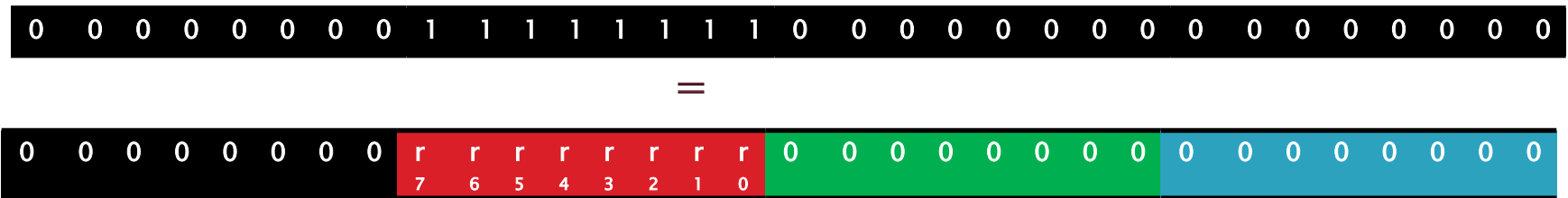
Image Organization

Pixel Representation as 32Bit Integer

Bit distribution:



Access through bit operation: **&**



Followed by a shift of 16 Bits towards right: >>16



```
// in Java
int c ...; // Pixel
int r = (c & 0x00ff0000) >> 16;
```

Table of Contents

Image

- 1 Basics
- 2 Image Resolution
- 3 Image Organization
- 4 **Color reduction / quantization**
- 5 Image Formats

Color reduction / Color quantization

- ▶ Color Resolution
 - Trade-off between quantization error and storage costs
 - Dependent on the representation capabilities of a device
- ▶ Different Approaches
 - Direct assignement (e.g. 32bit in some Color Space)
 - Color Lookup Table
 - Dithering

Quantization of grey-levels images

8 Bit

4 Bit

2 Bit

1 Bit



Color depth – Number of Bits pro Pixel (bpp)

Bits	number of colors		Name
1	2	(black + white)	bitonal
8	256	black to white	grey level
8	256	256 colors	Palette
16	65536	black to white	16 Bit grey levels
16	65536	almost all colors	High Color
24	16,7 Mio	all colors	24 Bit TrueColor
32	16,7 Mio	all colors	24 Bit TrueColor + Alpha
48	281 Bio	even more colors	48 Bit TrueColor

Indexed colors / color tables

- ▶ Instead of storing (r,g,b) for each pixel, only the Index of the pixel in a color **table is kept** „**colour lookup table**“ (CLUT)
 - The index has thus a small value (normally stored on one byte)
 - The **CLUT** (Palette) contains up to 256 (for 1-Byte Index) 24-bit values (often directly implemented in hardware (graphic card))
- ▶ The index value stored in the table is used to represent the color of a pixel

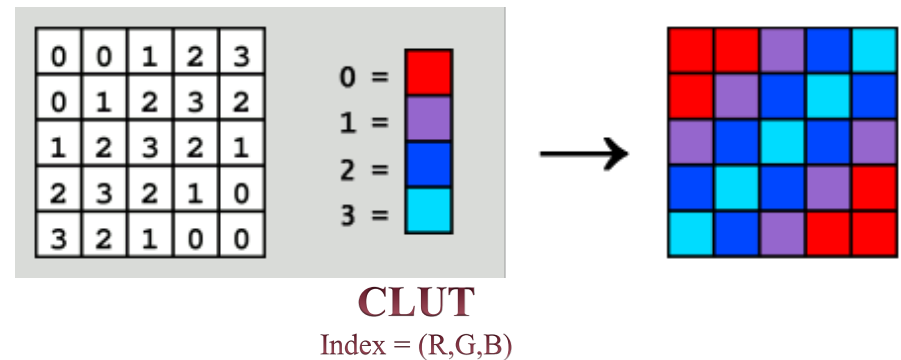
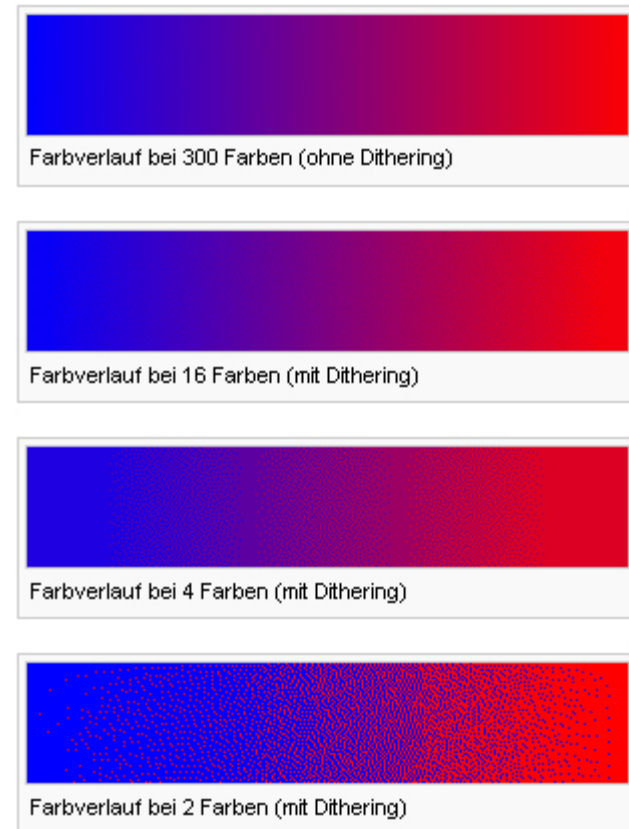


Image Source Wikipedia: https://en.wikipedia.org/wiki/Indexed_color

Color tables

- ▶ CLUT ideally contains the 256 **most important colors** of the image
- ▶ When an image gets **reduced** from 16.777.216 colors to the 256 most important ones, normally **not all colors** are found in the table
- ▶ Replacing the missing colors by the closest ones in the CLUT may cause a „**posterization**“ (Banding) effect
- ▶ **Dithering**: uses a pixel pattern and an optical shuffling
- ▶ **Web-secure** colors: 216 colors, known to be reproducible on all platforms by all browsers



Quelle: [http://de.wikipedia.org/wiki/Dithering_\(Bildbearbeitung\)](http://de.wikipedia.org/wiki/Dithering_(Bildbearbeitung))

Dithering

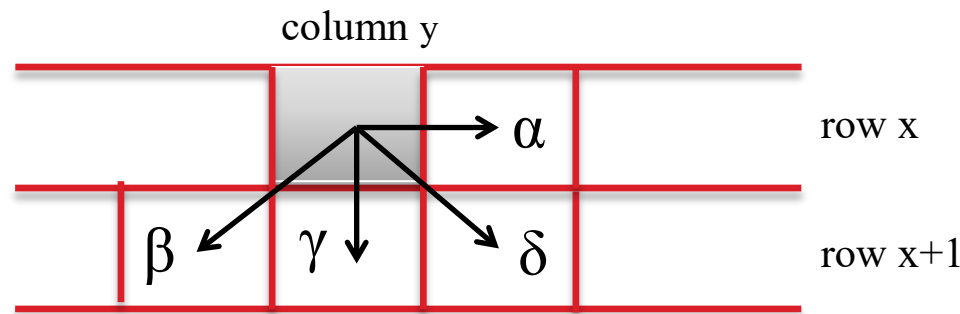
- ▶ **Apparent increase** of the number of perceivable colors (or levels) through spatial displacement
- ▶ **Types** of Dithering
 - Noise Dithering
 - Pattern Dithering
 - Error Diffusion (Floyd–Steinberg Algorithm)

Floyd–Steinberg Dithering I

- ▶ Published in 1976, widely used
- ▶ Distributes the quantization errors on neighbour pixels
 - The error is usually dispersed on the right and bottom pixels

- Distribution: $\frac{1}{16} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 7 \\ 3 & 5 & 1 \end{bmatrix}$

$$\alpha + \beta + \gamma + \delta = 1.0$$



Floyd Steinberg Dithering II

```
for (x=0; x<width; x++) {  
    for (y=0; y<height; y++) {  
  
        int oldpixel = pixels[x][y];  
        int newpixel = closestColor_in_CLUT(oldpixel);  
        int error = oldpixel - newpixel;  
        pixels[x][y] = newpixel;  
  
        pixels[x][y+1] += alpha*error;  
        pixels[x+1][y-1] += beta*error;  
        pixels[x+1][y] += gamma*error;  
        pixels[x+1][y+1] += delta*error;  
    }  
}
```

Dithering

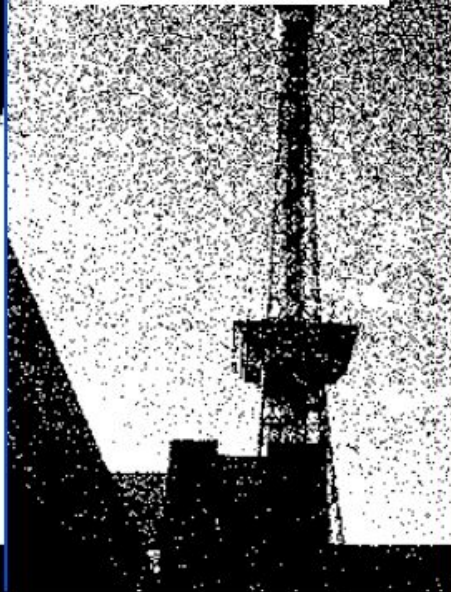
Example I



Schwellwert 128



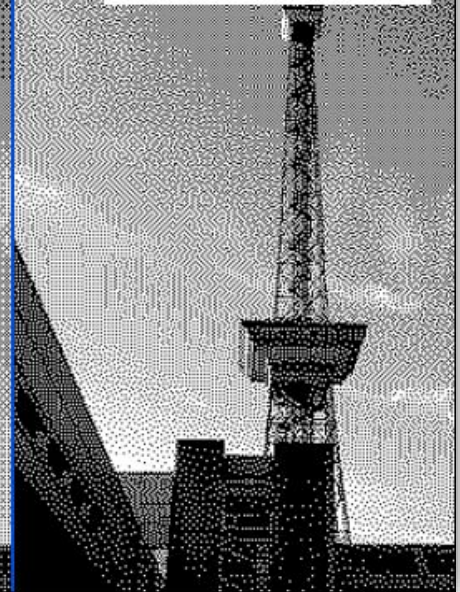
Noise Dithering



Pattern Dithering



Error Diffusion



Dithering

Example II



Original



20 colors



20 colors with Pattern dithering



Table of Contents

Image

- 1 Basics
- 2 Image Resolution
- 3 Image Organization
- 4 Color reduction / quantization
- 5 **Image Formats**

Image Formats

Choice of usual image formats

- ▶ **JPG:** JPEG File Format (1992)
- ▶ **TIFF:** Tagged Image File Format (~1985)
- ▶ **BMP:** MS Windows Bitmap
- ▶ **RLE:** Run-length encoded BMP
- ▶ **GIF:** Graphics Interchange Format (1987/9)
- ▶ **TGA:** Targa Image File (1984)
- ▶ **PNG:** Portable Network Graphics (1996)
- ▶ **PBM, PGM, PPM:** Portable Bit-/Grey-/Pix-Map (~1985)
- ▶ **RAW:** Raw data format (depends on producer)
- ▶ **PSD:** (Adobe) Photoshop Document
- ▶ **JP2, JPX, JPM:** JPEG2000 File formats (2002)
- ▶ ...

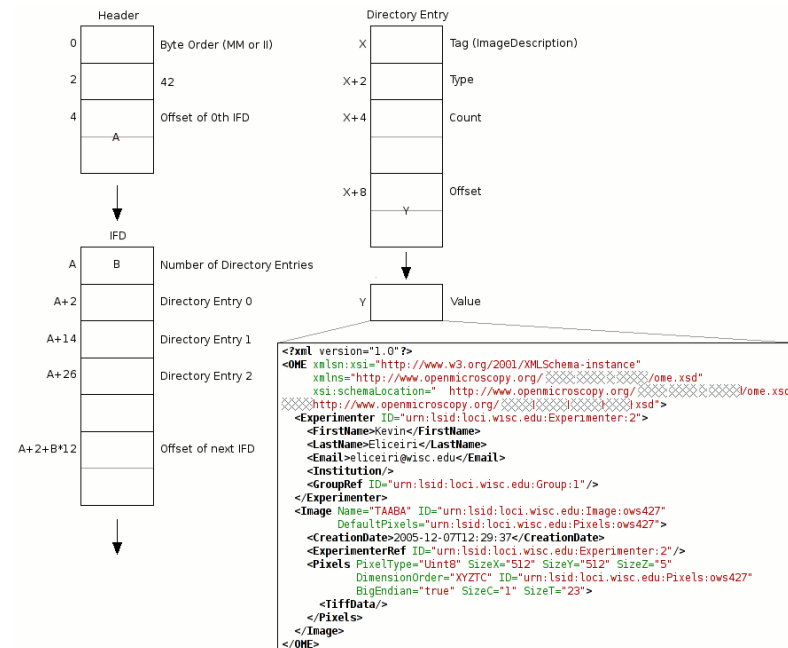
Properties of Image Formats

- ▶ Basic image parameters
- ▶ Metadata
- ▶ Functionalities / Interaction possibilities / Supported image types
- ▶ Type and principle of compression
- ▶ Support by browsers/software etc.
- ▶ Java Support
- ▶ Popularity / Interoperability

TIF-Format (Tagged Image File Format)

Developer: Aldus Corporation, Seattle, USA, 1986

- ▶ Very **flexible**, **high usage**, possibility of **compressed** storage
- ▶ Tiff-File: **composed of three main units**
 - **File header**:
 - First 8 Bytes.
 - 2 Bytes: **Number representation** (big or little endian),
 - 2 Bytes: **TIFF-Version number**
 - **File-Offset-pointer** to the first IFD-Block.
 - **Image-File-Directory** (e.g. one for each Resolution): 3 components:
 - **Number of Tag-Entries**
 - **Tag Array** (Array of 12 Bytes)
 - **File-Offset-pointer to next IFD**, or 0.
 - **Tags**:
 - Each Directory-Entry is called Tag; it consists of 12 Bytes. It contains **registration data and actual image information**.
 - 2 Bytes: **Tag-ID** (ca. 80 different ones, sorted in ascending order)
 - 2 Bytes: **Type** (SHORT, LONG, FLOAT, ASCII, BYTE)
 - 4 Bytes: **Length of data area**
 - 4 Bytes: **values**, if length of data area ≤ 4 (usually for metadata), or offset to data area



TIFF-data tags

► Main tags:

Description	Tag	Data types	Nb. values	Value
ImageWidth	100	short/long	1	...
ImageHeight	101	short/long	1	...
Bits per Pixel	102	short	1	...
Compression (1)	103	short	...	...
Color coding (2)	104	short	...	...
Origin (3)	112	short	...	...
Pointer to RGB color table	140	long	...	...
Information on used JPEG-coding	200 bis 209	short/long	...	...

- (1) options: no compression, CCITT-Formats (b/w Images), LZW (Lempel-Ziv-Welch), JPEG, Huffman Coding, etc.
- (2) options: WhitesZero, BlacksZero, b/w, RGB, RGB color table, CMYK (Cyan-Magenta-Yellow-Black), etc.
- (3) Image origin: up left, down right, down left, etc.

GIF-Format (Graphics Interchange Format)

- ▶ Developer: CompuServe Inc., Columbus, Ohio, USA, 1987
 - Several images may be stored in a single file
 - Goal: enable image exchange in a platform independent way
 - Characterized by „lossless“, efficient compression
 - Specification in 2 incompatible versions
- ▶ Parts of a GIF image:
 - Header:
 - Contains the GIF identification sequence and the version number of the used algorithm.
 - The end of the header indicates the beginning of the data area.
 - Application:
 - Enables encoding version and name information of the software used to produce or modify the image.

GIF-Format (Graphics Interchange Format)

- Trailer:
 - Marks the end of the GIF data streams.
- **Control:**
 - Controls the representation of the following image-block.
 - It is possible to specify **animation delay** as well as a transparent background color.
- **Image:**
 - Consists of an **Image Header**, an **optional color table** and of **pixel data**.
- **Comment:**
 - **Textual comment about an image block.**
- **Plain Text:**
 - Enables the **ASCII-based coding of texts inside an image** (font, color, size, position, stretching and direction or text).

Repeated for each
image

GIF-Format (Graphics Interchange Format)

- ▶ An image consists of
 - **Logical Screen Descriptor**
 - For coding the size, position and type of color table of the image,
 - **optional global or local color tables**
 - The global color table is used for all the following images except if they have an own local color table
 - The **pixel colors, noted as pointers to the color table**
- ▶ The pixel values are compressed by **LZW** (Lempel, Ziv, Welch)
 - Identifies repeating bit patterns of variable length in the data area → LZW-Analysis
 - Fills a table thanks to which the repeating patterns are replaced by shorter bit sequences (id of the sequence in the table)
 - **Frequent Bit patterns are replaced by shorter bit sequences**
- ▶ GIF is **only defined for Image Formats using an 8-bit-Color table**
 - ⇒ Cannot fulfill specific requirements on image quality

GIF – Summary

- ▶ Remarks:
 - Developed for fast transmission of Images under (then) low transmission rates.
 - **Image dimension** restricted to 16 000 x 16 000 pixels
 - **Number of colors** restricted to 256
 - The maximal color depth is thus 8 bits.
- ▶ Image coding:
 - **LZW**
compression rate: usually reaches 50 % (but 0 % is also possible).
- ▶ Summary:
 - **Very popular**, e.g. for inline-transmission of images on the Web
 - Cannot deal with 24-Bit Images (True-Color)

Portable Network Graphics (PNG)

- ▶ Graphics format for raster graphics
- ▶ Proposed as free and less complex than TIFF replacement for GIF, which was encumbered by patents until 2004
- ▶ Lossless data compression
- ▶ The compression may be improved using PNG prefilters:

Number	Name	Description
0	None	No prefiltering. Original pixels used
1	Sub	The differences to the next left neighbour are used
2	Up	The differences to the next top neighbour are used
3	Average	The differences to the average of the next top and left neighbours are used
4	Paeth	A so-called Paeth-Predictor, which uses differences to the next top left and diagonal top left neighbours, is applied.

Portable Network Graphics

- ▶ Pre-filters improve the compressibility of the data
- ▶ After pre-filtering: lossless deflate-Algorithm
 - Also used in
 - ZIP-Archive format (developed for it)
 - The gz-format of the archiving program gzip
 - The image format TIFF
- ▶ Deflate is currently the only supported method.
- ▶ The standard leaves however room for extensions
- ▶ The pre-filters enable a lower data size compared to GIF-files

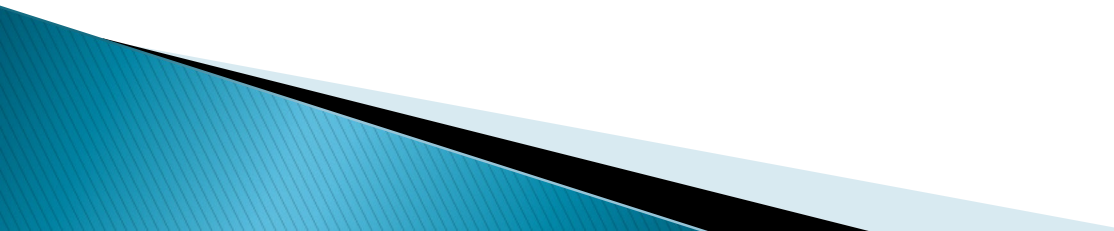
PNG Variants and Specifications – 1 –

PNG Variant	Color Depth / Features	Specification / Standard	Year Introduced	Typical Use
PNG-8	8-bit indexed color (up to 256 colors)	PNG Specification v1.0 — RFC 2083 / ISO/IEC 15948	1996	Icons, logos, simple graphics
PNG-24	24-bit true color (~16.7 million colors)	PNG Specification v1.0 — RFC 2083 / ISO/IEC 15948	1996	Photos, detailed graphics
PNG-32	24-bit color + 8-bit alpha transparency	Based on PNG truecolor + alpha channel (Color Type 6) in RFC 2083 / ISO/IEC 15948	~1996–1998	Transparent graphics, UI elements
APNG	Animation support for PNG images	Mozilla APNG Specification	2004	Animated web graphics
Interlaced PNG	Progressive image loading (Adam7 interlacing)	PNG Specification v1.0 — RFC 2083 / ISO/IEC 15948	1996	Web delivery over slow connections
Gray scale PNG	Grayscale image storage	PNG Specification v1.0 — RFC 2083 / ISO/IEC 15948	1996	Monochrome images, scans

PNG Variants and Specifications –2–

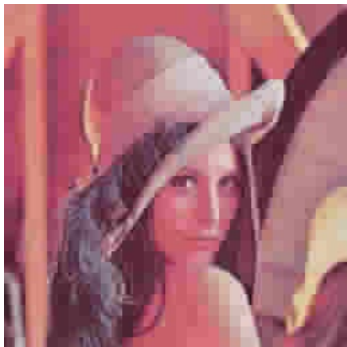
Standard	Description	Year
RFC 2083	Original PNG Internet specification	1997
W3C PNG Recommendation	Web standard recommendation by W3C	1996
ISO/IEC 15948	International standardized PNG specification	2003

JPEG

- ▶ Joint Photographic Experts Group
 - ▶ Committee created in 1986 by:
International Organization for Standardization (ISO)
and International Telecommunication Union (ITU)
 - ▶ A working group which creates the standard for still image compression
- 

JPEG – JPEG2000

- ▶ Two different standards
- ▶ Major differences :



JPEG

Created for natural imagery

Discrete Cosine Transform algorithm



JPEG2000

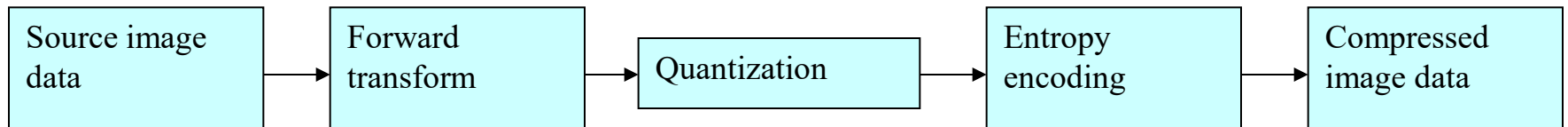
Created for computer generated imagery

Discrete Wavelet Transform algorithm

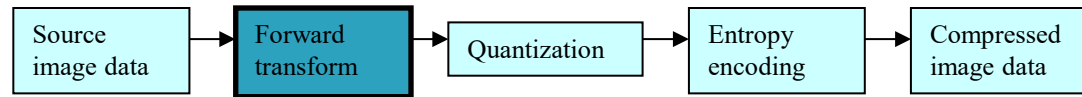
JPEG

► Architecture of standard

- Forward transform
- Quantization
- Entropy coding
 - DC coefficient
 - AC coefficients



JPEG



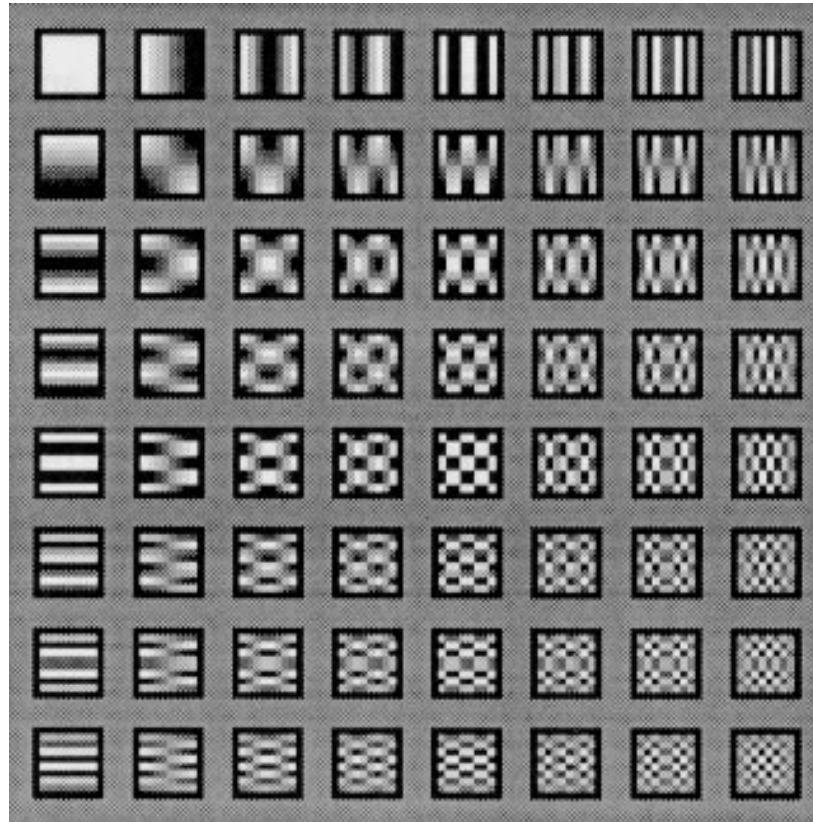
► Forward transform

1. Level shifted 2^{p-1}
2. Divided into blocks of 8×8 pixels
3. Forward DCT:

$$[C]_{i,j} = \begin{cases} \sqrt{\frac{1}{N}} \cos\left(\frac{(2j+1)i\pi}{2N}\right) & i = 0, j = 0, 1, \dots, N-1 \\ \sqrt{\frac{2}{N}} \cos\left(\frac{(2j+1)i\pi}{2N}\right) & i = 0, 1, \dots, N-1, j = 0, 1, \dots, N-1 \end{cases}$$

JPEG

- ▶ DCT basis functions



JPEG

- ▶ Original image

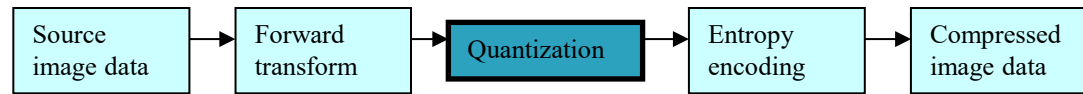
124	125	122	120	122	119	117	118
121	121	120	119	119	120	120	118
126	124	123	122	121	121	120	120
124	124	125	125	126	125	124	124
127	127	128	129	130	128	127	125
143	142	143	142	140	139	139	139
150	148	152	152	152	152	150	151
156	159	158	155	158	158	157	156

JPEG

► DCT coefficients

39.88	6.56	-2.24	1.22	-0.37	-1.08	0.79	1.13
-102.43	4.56	2.26	1.12	0.35	-0.63	-1.05	-0.48
37.77	1.31	1.77	0.25	-1.50	-2.21	-0.10	0.23
-5.67	2.24	-1.32	-0.81	1.41	0.22	-0.13	0.17
-3.37	-0.74	-1.75	0.77	-0.62	-2.65	-1.30	0.76
5.98	-0.13	-0.45	-0.77	1.99	-0.26	1.46	0.00
3.97	5.52	2.39	-0.55	-0.051	-0.84	-0.52	-0.13
-3.43	0.51	-1.07	0.87	0.96	0.09	0.33	0.01

JPEG

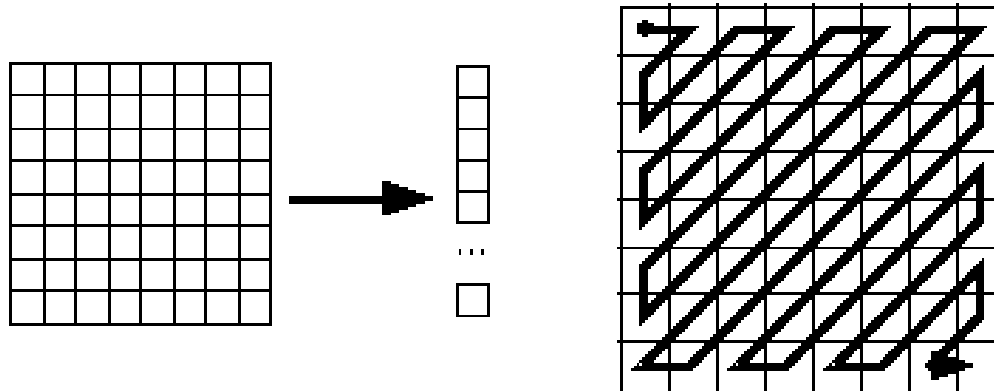


► Quantization

- Quantization table

$$l_{ij} = \left\lfloor \frac{\theta_{ij}}{Q_{ij}} + 0.5 \right\rfloor$$

- Zigzag scan

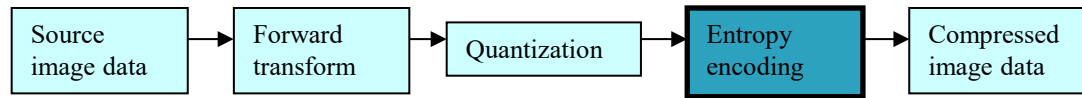


JPEG

- ▶ Quantization table

16	11	10	16	24	40	51	61
12	12	14	19	26	58	60	55
14	13	16	24	40	57	69	56
14	17	22	29	51	87	80	62
18	22	37	56	68	109	103	77
24	35	55	64	81	104	113	92
49	64	78	87	103	121	120	101
72	92	95	98	112	100	103	99

JPEG



► Entropy encoding

- DC coefficient and AC coefficients are coded differently
- DC coefficient
 - Encode the difference
 - Huffman code unmanageable

JPEG

- ▶ More accurate image
 - Increase the bit rate
 - Decrease quantization table elements
- ▶ Less accurate image
 - Decrease the bit rate
 - Increase quantization table elements
- ▶ Problem:

Low rate image: « blockiness »

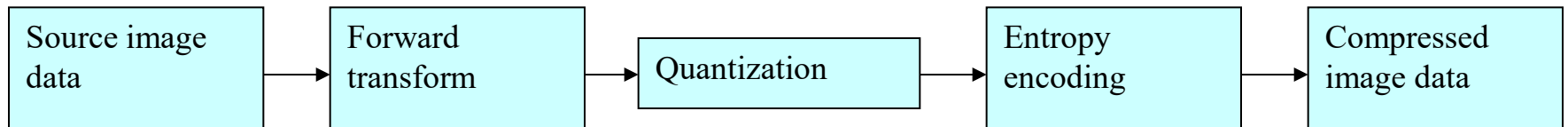
JPEG2000

- ▶ Can be both lossless and lossy
- ▶ Improves image quality
- ▶ Uses a layered file structure :
 - Progressive transmission
 - Progressive rendering
- ▶ File structure flexibility:
 - Could use for a variety of applications
- ▶ Many functionalities

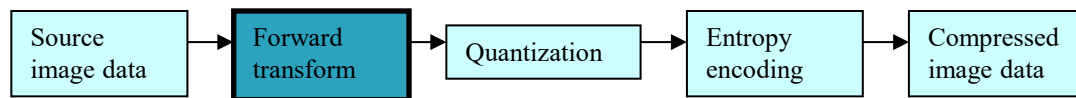
JPEG2000

► Architecture of standard

- Forward transform
- Quantization
- Entropy coding



JPEG2000

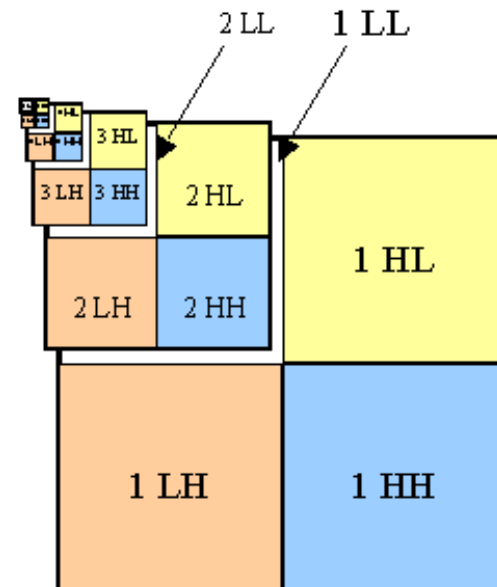
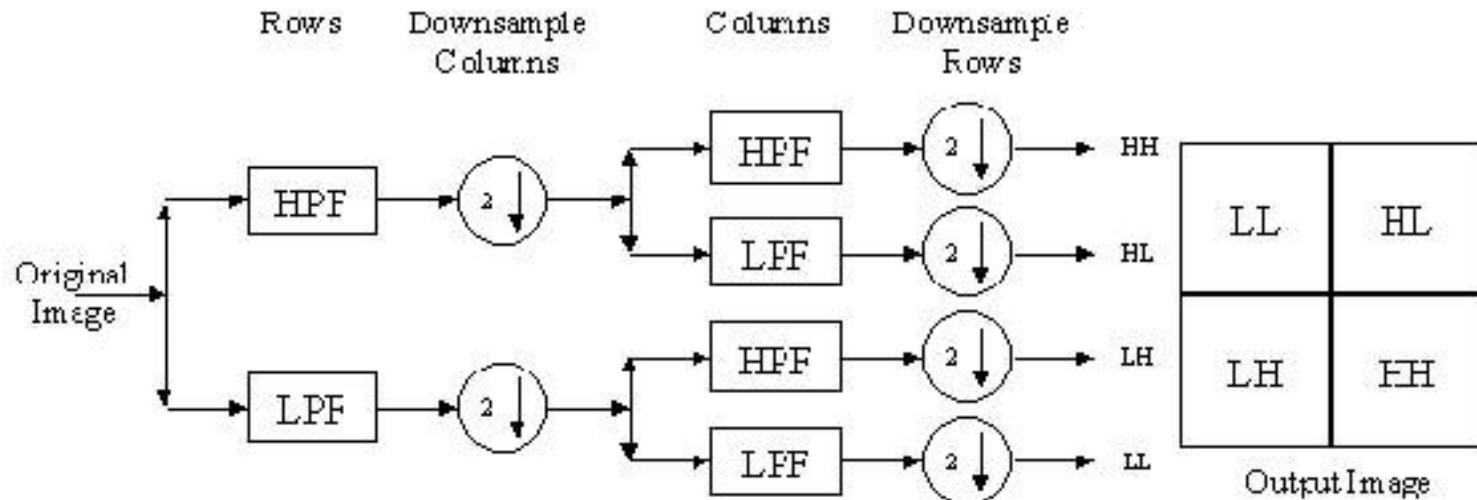


► Forward transform

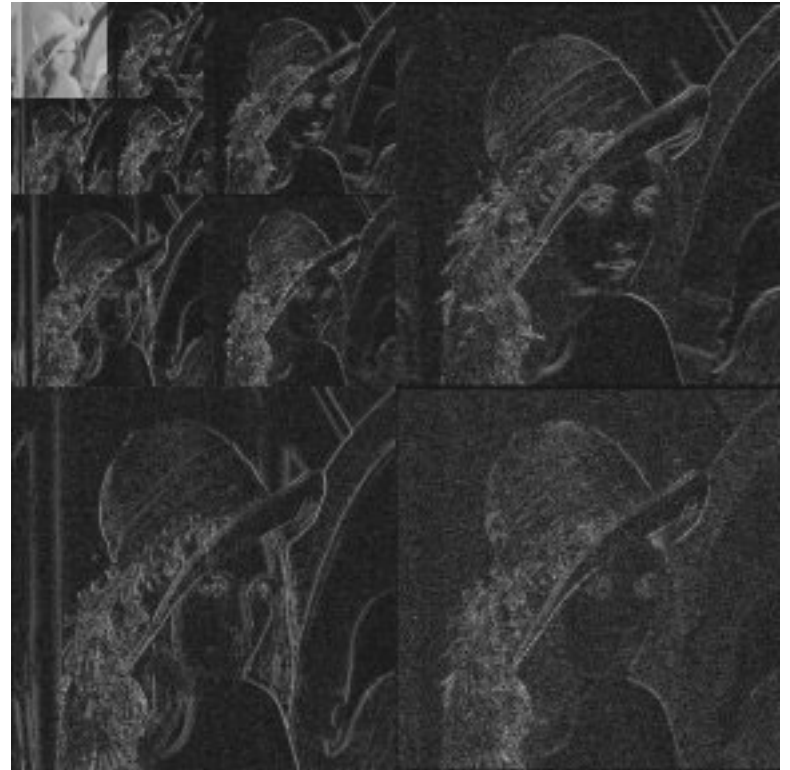
1. Level shifted
2. Divided into blocks
3. Forward Discrete Wavelet Transform

JPEG2000

► Details



JPEG2000



Which format for which usage?

Format	Storage req.	Application	Remark
JPEG	Low to average	P, D, W	Lossless compression
TIFF	hoch	M, I, P, D	Lossless compression possible
GIF	Average	M, I, W	Indexed colors, possible high loss of tones
PNG	Average (or low)	M, I, W, (P)	Lossless compression possible

Anwendungsbereiche: M = Monochrome images,
I = Illustrations,
P = Photos,
W = Images on WWW,
D = Data exchange with graphic companies

[But: to be taken with a pinch of salt!]

The End